Reinventing Primality Testing through Randomized Algorithms

CS648 Project Presentation

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Indian Institute of Technology Kanpur

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Guided by Prof. Surender Baswana

Outline

1. Introduction & Motivation
2. Deterministic Algorithms
3. Fermat Primality Test
4. Miller–Rabin Test
5. Experimental Results
6. RSA Application & Error Contextualization
7. Comparison & Conclusion

Why Primality Testing?

The Core Problem

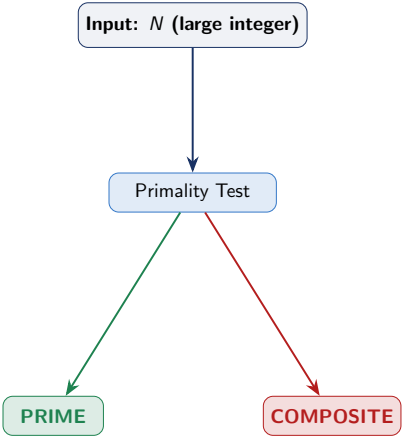
Given a number N, determine whether it is **prime** or **composite**.

Why does it matter?

- Foundation of **RSA** and **Diffie–Hellman** cryptography
- Secure communication needs large primes (1024–4096 bits)
- Must test *thousands* of candidates to find each prime

The Challenge

Deterministic methods are provably correct but *too slow* for large inputs.



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Trial Division

Check all $d \in [2, \lfloor \sqrt{N} \rfloor]$

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$\Rightarrow$ **Randomized algorithms** dominate in practice.

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If $a^{n-1} \not\equiv 1 \pmod{n} \Rightarrow n$ is **definitely composite**.

Partitioning the Search Space

Partition $A = \{1, \ldots, n-1\}$ into:

- $A' = \{a \mid \gcd(a, n) = 1\}$ *(units mod n)*
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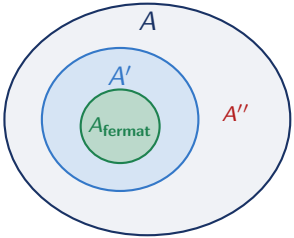

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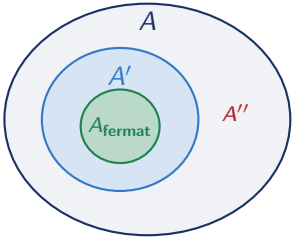

Proof of half-density:

Let $x \in A' \setminus A_{\text{fermat}}$. The set xL (where $L = A_{\text{fermat}}$) is disjoint from L and has the same size. Since both are subsets of A':

$$|L| + |xL| \leq |A'| \implies |A_{\text{fermat}}| \leq \frac{|A'|}{2}$$

Carmichael Numbers — The Blind Spot

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A composite n where $A' \subseteq A_{\text{fermat}}$:
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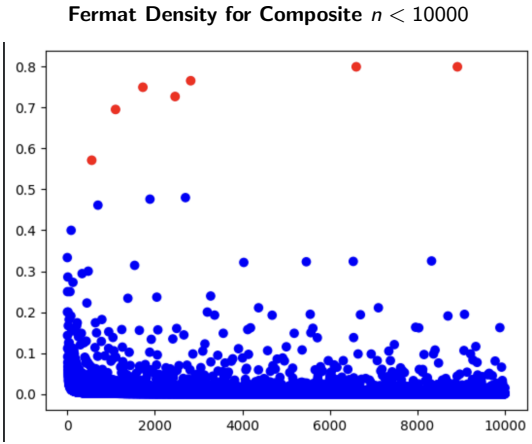
The Fermat test is fundamentally broken for Carmichael numbers.



Fermat Test — Algorithm & Error Analysis

Algorithm:
`Fermat_Primality_Test`(n, k)

```
if n is even then return COMPOSITE
end if
for i = 1 to k do
    Pick random a ∈ [2, n − 2]
    g ← gcd(a, n)
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Modular exponentiation dominates.

The Key Insight — Square Roots of Unity

Theorem (Nontrivial Square Roots)

If n is an odd prime and $x^2 \equiv 1 \pmod{n}$, then $x \equiv 1 \pmod{n}$ or $x \equiv n-1 \pmod{n}$.

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Miller–Rabin's Improvement

Write $n - 1 = 2^s \cdot d$ with d odd. Inspect the squaring sequence:

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- A violation $\Rightarrow$ **non-trivial root of 1** $\Rightarrow$ **composite**

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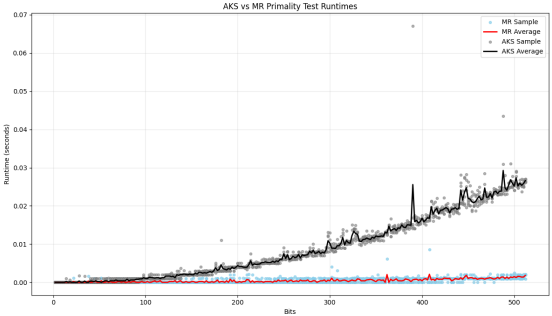
- Stronger per-round bound ($\frac{1}{4}$ vs $\frac{1}{2}$)
- Same asymptotic complexity

AKS vs Miller–Rabin: Runtime Comparison



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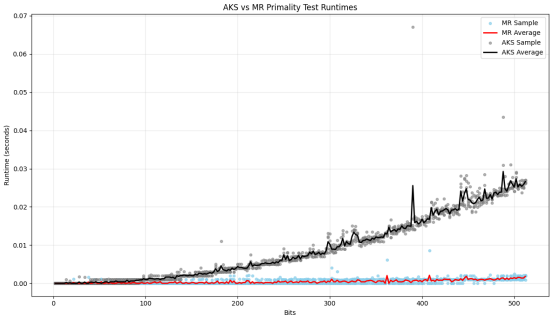

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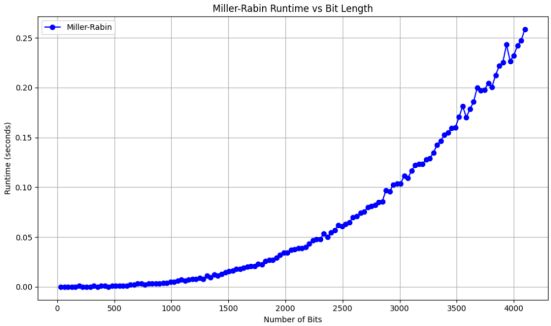

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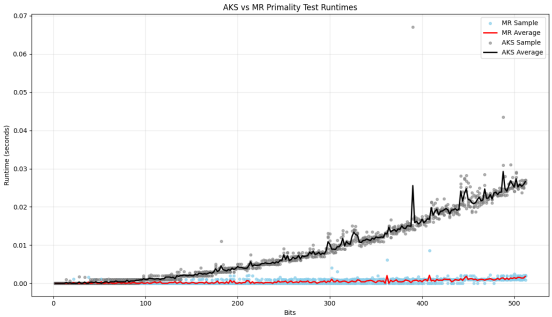

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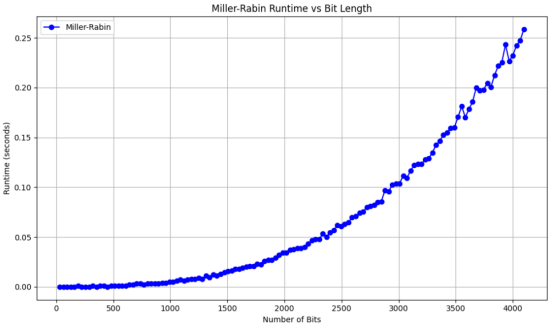

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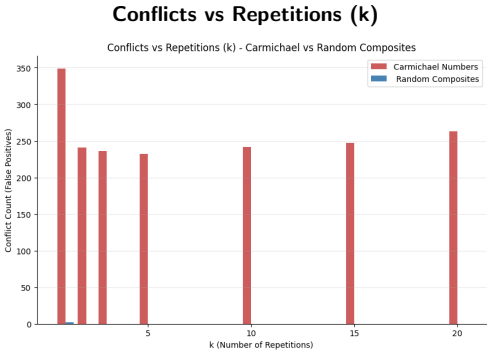
Key takeaway: Miller-Rabin scales to cryptographic sizes; AKS does not.

False Positives: Carmichael vs Random Composites



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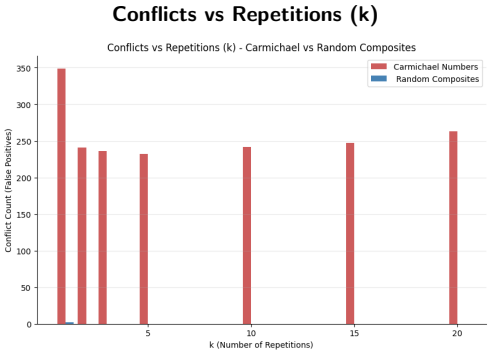

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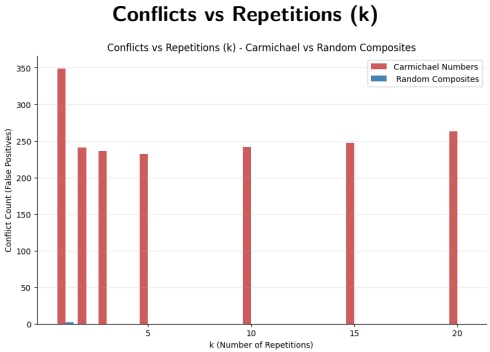

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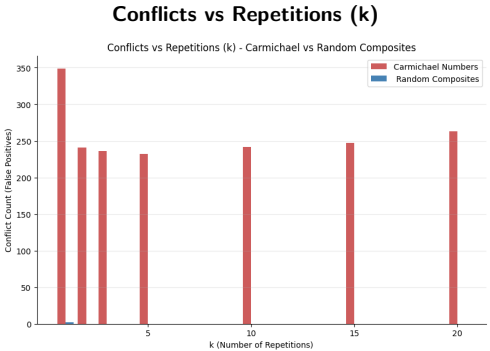

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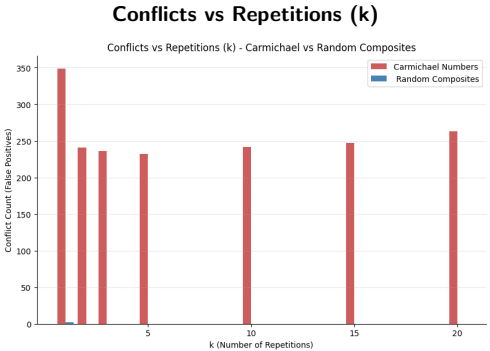

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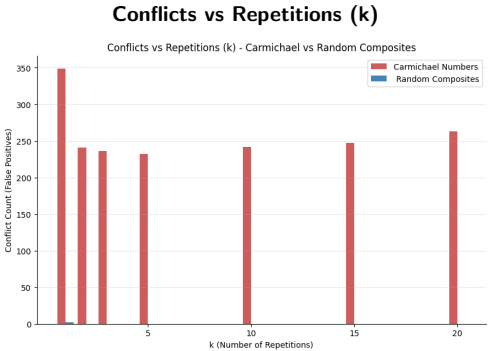

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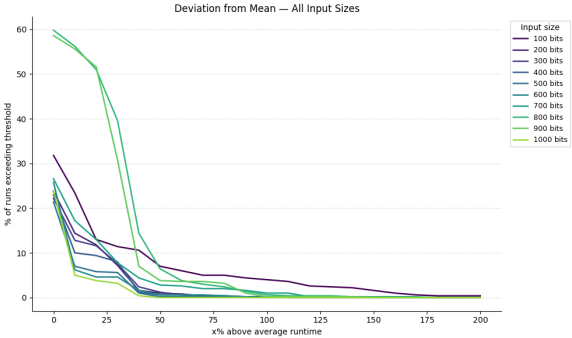
Why Miller-Rabin Wins

Nontrivial square root witnesses are abundant even for Carmichael numbers.

Runtime Consistency — Deviation Analysis

Each input size benchmarked **2000 times**.
Measured: % of runs exceeding mean by at least x%.



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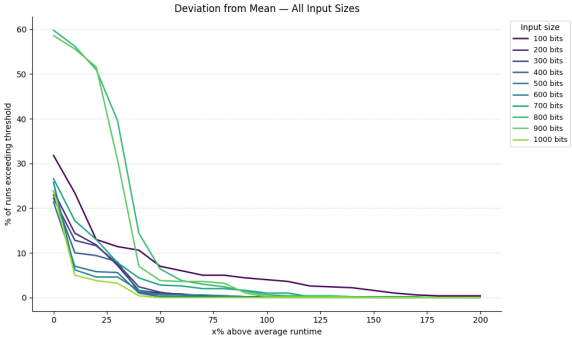

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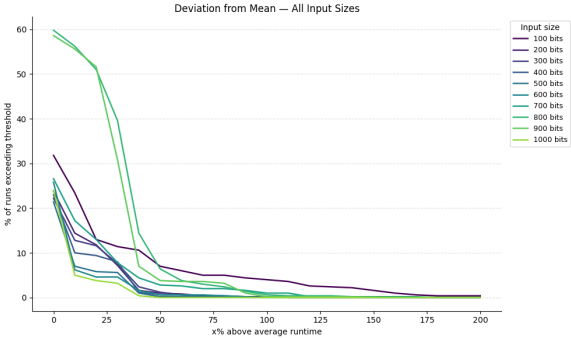

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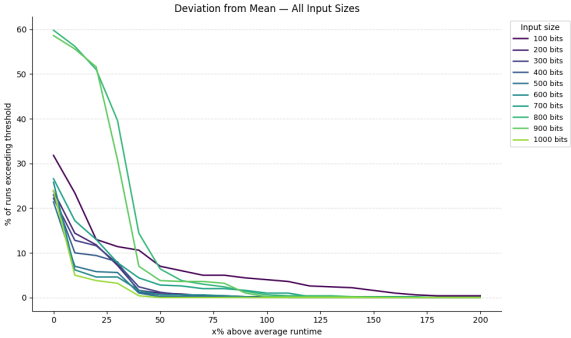

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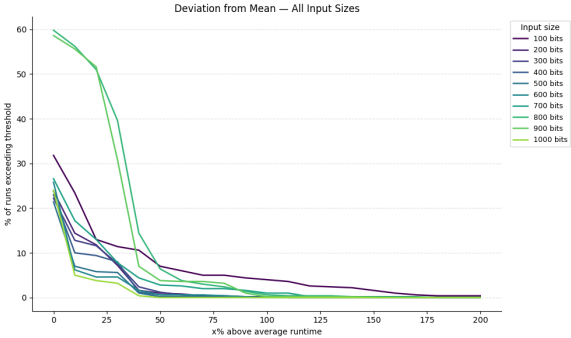

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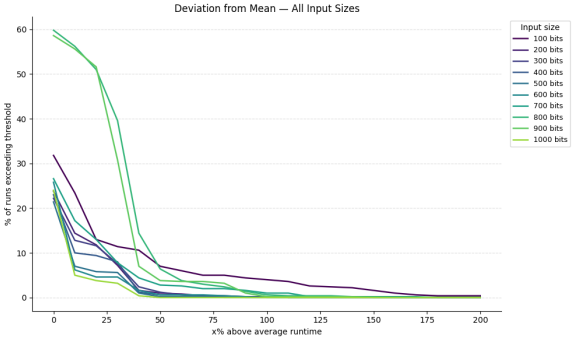

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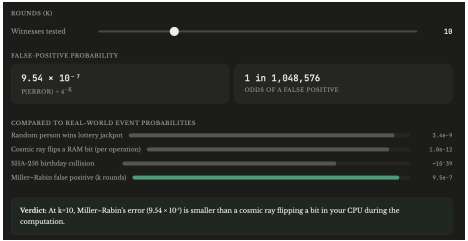
Conclusion

Miller-Rabin is **highly predictable** at cryptographic scales.

Probabilistic Error in Context

Error Probability Decay (4^{-k})



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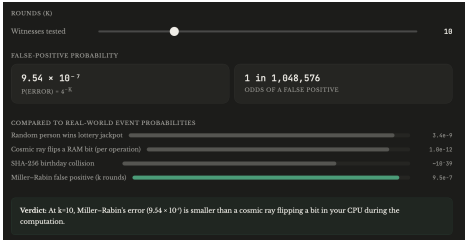

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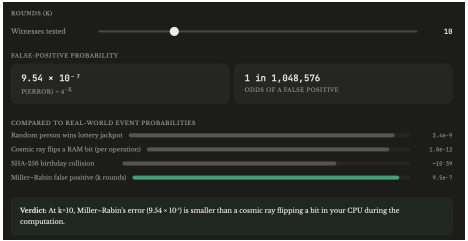

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Practical Determinism

At $k = 20$, Miller-Rabin error is on par with *hardware failure*. At $k = 40$, it is considered **cryptographically safe**.

RSA Key Generation — Practical Impact

RSA-1024 key generation requires:

- Two 512-bit primes p and q
- By the Prime Number Theorem, ≈ 1 in 355 candidates is prime
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Miller-Rabin

Probabilistic yet *effectively deterministic* at $k \geq 20$. The de-facto standard in modern cryptography.

Conclusion

Key Takeaways

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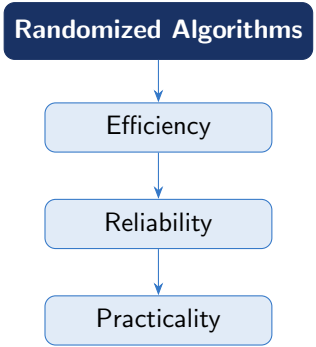

Thank You

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